

Jiahao Cao

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MPhil 2028 · AI Agents & Reinforcement Learning



EDUCATION

The Chinese University of Hong Kong, Shenzhen · MPhil in Artificial Intelligence 2026.09 - 2028.06

- Supervisor: Menglin Liu | Research: AI Agents and Reinforcement Learning

Xi'an Jiaotong University · B.Eng. in Software Engineering 2022.09 - 2026.06

- GPA: 85.78/100 | Selected Coursework: Natural Language Processing (96), Operating Systems (93), Algorithms (91)
- Thesis: Built a prompt-triggered backdoor attack system for text-to-image diffusion models, optimized for attack success rate.

National University of Singapore · School of Computing Summer Program (Grade: A) 2024.07 - 2024.08

- Grade: A | Focus: Artificial Intelligence, Deep Learning, and FinTech
- Capstone: Built an intelligent financial analytics platform with LSTM/XGBoost for stock trend prediction.

TECHNICAL SKILLS & OPEN-SOURCE CONTRIBUTIONS

LLMs & Agents: Multi-agent Orchestration, LangGraph, MCP, RAG, Agent Memory, Context Engineering, Skills, LLM-as-a-Judge

Programming & ML: Python, PyTorch, FastAPI, JavaScript/TypeScript, Node.js, SQL, Git, Linux

Agent Development Tools: Claude Code, Codex, Pi

 OpenViking · Open-Source Agent Memory Contributor (37.7k Stars) 2026.04

- Merged an OpenCode plugin upstream (PR #1859) that captures session context, commits memory at lifecycle events, and retrieves/injects repository-indexed long-term memory via MCP.
- Built a Pi agent plugin reusing the same capture-commit-retrieve-inject workflow.
- Improved benchmark reproducibility through packaging, automated tests, cross-runtime debugging, and documentation.

RESEARCH EXPERIENCE

Intent-Driven Auto-Research Multi-Agent System · Research Intern 2026.03 - 2026.07

KDD Submission Under Review · Fourth Author

- Co-designed intent-aware AutoResearch: persistent method contracts span planning through writing, while evidence gates and an independent Critic constrain analysis and claims, adapting methods across analytics, empirical studies, and reviews.
- Led an LLM-as-a-Judge plus rule-based benchmark across **48 real-domain tasks** (21 industry / 18 policy / 9 empirical) and **8 quality dimensions**; versus single-agent and general-research baselines, raised **factual accuracy by 32%** and achieved **94% evidence-supported conclusions**.

PROJECT EXPERIENCE

SLRHarness · Automated Literature Review Pipeline 2026.08

- Designed an artifact-first Claude Code pipeline for scope, multi-source retrieval, metadata, synthesis, and audit; resumable recovery cut full rerun overhead by **87%**, with zero redundant work on completed stages.
- Built a global registry that deduplicates cross-topic evidence by DOI, arXiv ID, title, and authors; across reviews covering **120+ papers**, achieved **96% duplicate-source retrieval precision** and **<3% miss rate** on core papers.
- Enforced deterministic output contracts and validation rules, achieving **92% compliance** across 5 core artifacts: synthesis, source list, metadata, audit report, and references.

Feishu LongMemory · Evidence-First Personal Agent Memory 2026.05

- Built FastAPI/SQLite long-term memory for Feishu/OpenClaw Agents, covering ingestion, evidence-grounded extraction, profiles, context injection, auditable forgetting, and reminders; duplicates converge to one canonical memory with a full evidence chain.
- Designed an LLM-candidate, multilayer deterministic-validation, and rule-fallback pipeline, reaching **92.7% memory-extraction accuracy** and **90% interception** of sensitive data such as keys, contact details, and private keys.
- Implemented task-aware retrieval for collaboration, weekly reports, and reminders, achieving **91% Top-3 relevant-memory recall**; memory injection improved Agent response-personalization consistency by **58%**.

HONORS & AWARDS

Provincial First Prize · Chinese Collegiate Computing Competition (Financial AI) 2025

Provincial Third Prize · Lanqiao Cup (Java) 2024 Third-Class Scholarship · Xi'an Jiaotong University 2023-2025

Provincial Third Prize · Chinese Mathematics Competitions 2024 Outstanding Student · Xi'an Jiaotong University 2024-2025